

## M624 HOMEWORK – SPRING 2026

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SET 1 - DUE 02/17/2026

**From Chapter 4 :**

**Additional Problem 1:** Prove the Auxiliary Lemma we stated in Lecture 2. Namely prove the following:

A normed vector space  $(X, \|\cdot\|)$  is *complete* if and only if every absolutely convergent series in  $X$  converges in  $X$ .

Recall; i) A series  $\{x_n\}_n$  is said to be absolutely convergent if  $\sum_{n=1}^{\infty} \|x_n\| < \infty$ .

ii) A series  $\sum_{n=1}^{\infty} x_n$  converges in  $X$  if there exists an  $x$  in  $X$  such that  $\lim_{N \rightarrow \infty} \sum_{n=1}^N x_n = x$  in  $X$ ; that is if  $\|\sum_{n=1}^N x_n - x\| \rightarrow 0$  as  $N \rightarrow \infty$

**Additional Problem 2:** For any  $1 \leq p < \infty$  consider the space

$$L^p(\mathbb{R}^d) := \{f : \mathbb{R}^d \rightarrow \mathbb{C}, \text{ measurable, } \|f\|_{L^p(\mathbb{R}^d)} := \left( \int_{\mathbb{R}^d} |f(x)|^p dm \right)^{\frac{1}{p}} < \infty\}.$$

Assume that  $\|f\|_{L^p(\mathbb{R}^d)}$  is a norm<sup>†</sup> whence  $d_p(f, g) := \|f - g\|_{L^p(\mathbb{R}^d)}$  defines a metric and  $L^p$  is a metric space. Use the proof of completeness that I gave in class from Folland (Lecture 3) when  $p = 2$  to **prove** that  $L^p(\mathbb{R}^d)$  is *complete*.

(†) Question: can you guess what would you need to prove the triangle inequality when  $p \neq 2$ ? Justify your answer.

**From Chapter 4 (pp 193-194):** 1, 2\*, 3, 4 (show completeness only), 5, 6a), 7, 8a), 10.

\* to show that  $f - g$  is *orthogonal* to  $g$  you need to show that  $\langle f - g, g \rangle = 0$ .